Reg. No.	R	A	2	3	1	1	0	0	3	0	1	1	3	5	0

B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2024

First & Second Semester

21GNH101J – PHILOSOPHY OF ENGINEERING

(For the candidates admitted from the academic year 2022-2023 onwards)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours Max. Marks: 75

PART – A (20 × 1 = 20Marks) Marks BL CO PO

Answer **ALL** Questions

1. _____ refers to the elements of arts that are independent of its interpretation of significance — 1 1 1 1
   (A) Art form (B) Arts and science
   (C) Engineering (D) Philosophy

2. Art therapy and the diagnostic drawing series are examples of art used for — 1 2 1 1
   (A) Entertainment (B) Expression of imagination
   (C) Bringing about political change (D) Raising awareness for social causes

3. The _____ mechanism is the earliest known model of a mechanical computer in history. — 1 2 1 1
   (A) Antikythera (B) Pneumatic
   (C) Phaistos (D) Cosmos

4. The first Electrical Engineer is — 1 2 1 1
   (A) De Magnete (B) James Maxwell
   (C) Thomas Savery (D) William Gilbert

5. The first stage of product life cycle is — 1 1 2 1
   (A) Product decline (B) Product growth
   (C) Product development (D) Product maturity

6. _____ is natural extension of PLM and creates a truly full life cycle that takes your obsolete or used products back into raw materials. — 1 2 2 1
   (A) Half loop cycle (B) Semi loop cycle
   (C) Open loop cycle (D) Closed loop cycle

7. Choose a relevant ontology which suits for a specific domain independent criteria — 1 1 2 1
   (A) Functional (B) Reference
   (C) Foundational (D) Domain

8. When a product is in its decline stage of its product life cycle the average selling price will be — 1 1 2 1
   (A) Higher (B) Average
   (C) As decided by manufacturer (D) Lower

9. "RAISEC" model was theorized by — 1 1 3 1,4
   (A) Isaac Newton (B) Imhotep
   (C) Archimedes (D) John Holland

10. _____ is creating new tools and devices. — 1 1 3 1,4
   (A) Engineering (B) Science
   (C) Physics (D) Chemistry

11. In this dimension of engineering, the completed job, which stands before the world, leads to higher recognition — 1 2 3 1,4
   (A) Basic sciences (B) Social sciences
   (C) Design (D) Practical accomplishment

12. Popper's concept of 'Critical Discussion' is related to — 1 2 3 1,4
   (A) Design as activity (B) Design as planning
   (C) Design as epistemology (D) Design as enterprising

13. _____ performs experiments using the scientific method — 1 1 4 1,5
   (A) Engineers (B) Doctors
   (C) Scientists (D) Students

14. _____ spin-offs or the variations of the "Addie Model" — 1 2 4 1,5
   (A) Addie model (B) Scientific model
   (C) Engineers model (D) Dick and Carey

15. _____ is the final stage of engineering design process — 1 2 4 1,5
   (A) Communicate results (B) Brainstorm possible solutions
   (C) Research ideas (D) Select an approach

16. The evaluation phase consist of _____ number of parts. — 1 1 4 1,5
   (A) 2 (B) 4
   (C) 6 (D) 8

17. _____ are the reason for the phenomenal growth in technology of every generation. — 1 2 5 1,9
   (A) Scientists (B) Engineers
   (C) Actors (D) Doctors

18. The International Engineering Consortium was established in — 1 2 5 1,9
   (A) 1922 (B) 1944
   (C) 1966 (D) 1988

19.	The _____ is noted for being the worlds largest technical professional organization (A) National Society of Professional Engineers (B) IEEE (C) American association of Engineering Societies (D) Society of Women Engineers	1	1	5	1,9
20.	Sustainability is concerned about our (A) Environment (B) Companies (C) Resources (D) Society	1	1	5	1,9

PART – B (4 × 10 = 40 Marks)
Answer **ANY FOUR** Questions

		Marks	BL	CO	PO
21.	Write short notes on steam pyramid with a neat sketch.	10	2	1	1,6
22.	What is product life cycle? Explain.	10	2	2	1,6
23.	Explain in detail the four dimensions of Engineering in detail.	10	2	3	1,4
24.	Explain the design phase and development phase in ADDIE model.	10	2	4	1,5
25.	Write short notes on CDIO Engineers in industry.	10	2	4	1,5
26.	Write short notes on Society of Women Engineers.	10	2	5	1,9

PART – C (1 × 15 = 15 Marks)
Answer **ANY ONE** Question

		Marks	BL	CO	PO
27.	Give detailed notes on RAISEC model.	15	2	2	1,4
28.	Classify the desired attributed of an Engineer with a minimum of ten key points.	15	2	1	1,4

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